

SMART ENERGY CONSUMPTION OPTIMIZER USING AI

Team Member

Bharani.S

Bharathi.M

Boopathi.P

Gayathri.P.M

Content

- Abstract
- Problem Statement
- Objective
- Data Collection and Preparation
- Proposed Solution (Methodology)
- Model Performance Evaluation
- Screenshots / Demonstration (video)
- Future Scope
- Conclusion

Abstract

This project aims to optimize household or industrial energy usage through AI-driven predictive analytics. By analyzing real-time and historical consumption data, the system forecasts peak demand, detects wasteful patterns, and automatically recommends or executes energy-saving strategies. This not only reduces costs but also contributes to sustainable energy usage.

Problem Statement

Traditional energy consumption is often inefficient due to lack of real-time monitoring and intelligent analysis. Users lack actionable insights into usage patterns, leading to high bills and energy waste. A smart AI-based system is needed to optimize consumption dynamically.

Objective

- Monitor and analyze real-time energy usage.
- Predict high-energy consumption periods.
- Suggest or automate strategies to reduce usage.
- Support integration with smart appliances and grid systems.
- Help users lower energy bills and carbon footprint.

Data Collection and Preparation

- Sources: Smart meters, IoT sensors, weather data, user behavior logs.
- Features: Time of day, appliance usage, temperature, occupancy.
- Preprocessing: Cleaning missing values, normalizing readings, feature engineering.
- Optional: Use open datasets like UCI's Individual household electric power consumption dataset.

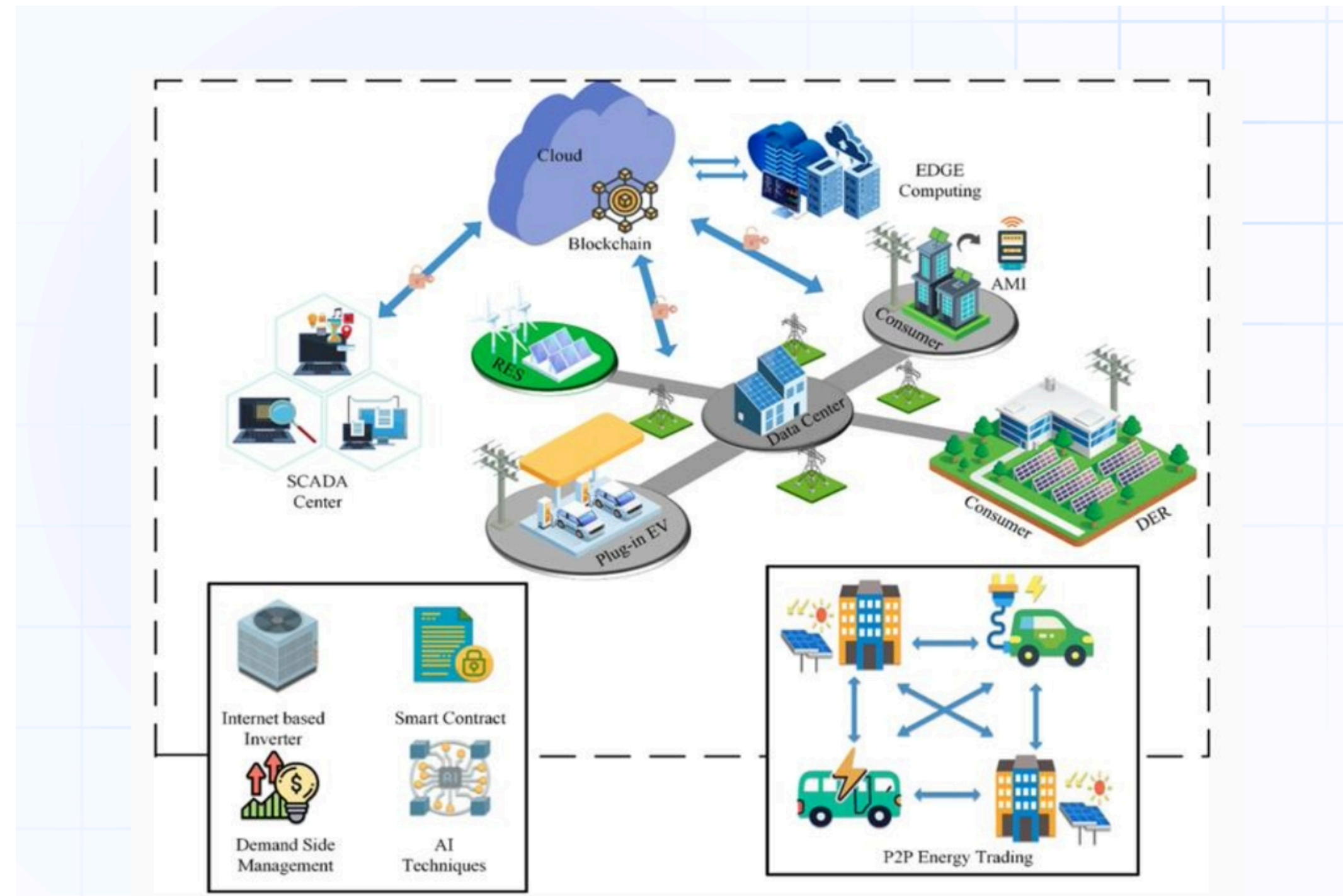
Proposed Solution (Methodology)

1. Data Collection Module: Fetch real-time and historical data.
2. AI Model:
 - Time series forecasting (LSTM/Prophet) for demand prediction.
 - Clustering (K-Means) for user segmentation.
 - Classification (Random Forest/SVM) for anomaly detection.
3. Optimization:
 - Rule-based automation or reinforcement learning to suggest optimal usage schedules.
4. User Interface: Dashboard showing live stats, alerts, and suggestions.

Model Performance Evaluation

- Accuracy: Prediction vs actual usage (MAE, RMSE).
- Efficiency: Energy savings in percentage.
- Response time: Real-time adaptability.
- User feedback and satisfaction surveys.

Screenshots / Demonstration (video)



Future Scope

- Integration with smart grid for demand-response.
- Use of edge computing for offline optimization.
- Expand to community-level optimization.
- Integration with renewable energy predictions.

Conclusion

AI can significantly enhance energy consumption efficiency by identifying patterns, predicting demand, and recommending intelligent optimizations. The solution provides economic and environmental benefits while encouraging sustainable energy practices.